Gradient tones obviate floating features in Oku tone sandhi

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Main Claim

- Hyman (2014) posits several **floating tones** in an analysis on Tonal Sandhi in Oku to block or trigger certain processes.
- We present an **alternative** account without floating tones, based on the intuition of weak and strong tones formalized as **gradient** representations (Goldrick and Smolensky 2016).

Data

Data: Oku Tones

- Grassfields Bantu, North West of Cameroon

(1) Four Level tones

a.	īkón	‘bean’	H
b.	ndəi	‘cloth’	M
c.	njàm°	‘axe’	L°
d.	kà	‘basket’	L

(2) Four Contour tones

a.	ɲâm	‘animal’	HL
b.	jím̄sē	‘backs’	HM
c.	ntō` k	‘bundle’	ML
d.	ncìí	‘chief’s house’	LH

Data: Oku Noun Plural

- Both the plural suffix and the noun stem can change tones when they are concatenated.

(3) Tone Sandhi in Oku plural forms

	SG	PL		SG	PL
a.	kà	kà-**sə̀**°	‘basket(s)’	L	L.L°
b.	njàm°	njàm-**sé**	‘axe(s)’	L°	L.H
c.	ndəi	ndəi-**sé**	‘cloth(s)’	M	M.H
d.	səə	səə-**sé**	‘fish’	M	H.H
e.	yúo	yúo-**sé**	‘bee(s)’	H	H.H
f.	ntō` k	ntōk-**sè**°	‘bundle(s)’	ML	M.L°
g.	nyâm	nyám-**sē**	‘animal(s) ’	HL	H.M

Data: Generalizations I

- The plural suffix bears
  - . . . an unreleased low tone after a low tone or an (underlying) mid low tone.
  - . . . a mid tone after an underlying falling tone.
  - . . . a high tone otherwise.

(4) Plural suffix tones in Oku

	SG	PL		SG	PL
a.	kà	kà-**sə̀**°	‘basket(s)’	L	L.L°
b.	ntō` k	ntōk-**sè**°	‘bundle(s)’	ML	M.L°
c.	nyâm	nyám-**sē**	‘animal(s) ’	HL	H.M

Data: Generalizations II

- On stems
  - . . . contours are simplified (general phonological process).
  - . . . unreleased low tones become low tones (general phonological process).
  - . . . some mid tones become high tones.

(5) Tonal stem changes in Oku plural forms

	SG	PL		SG	PL
a.	njàm°	njàm-**sé**	‘axe(s)’	L°	L.H
b.	ndəi	ndəi-**sé**	‘cloth(s)’	M	M.H
c.	səə	səə-**sé**	‘fish’	M	H.H
d.	ntō` k	ntōk-**sè**°	‘bundle(s)’	ML	M.L°
e.	nyâm	nyám-**sē**	‘animal(s) ’	HL	H.M

Background

Background: Assumptions

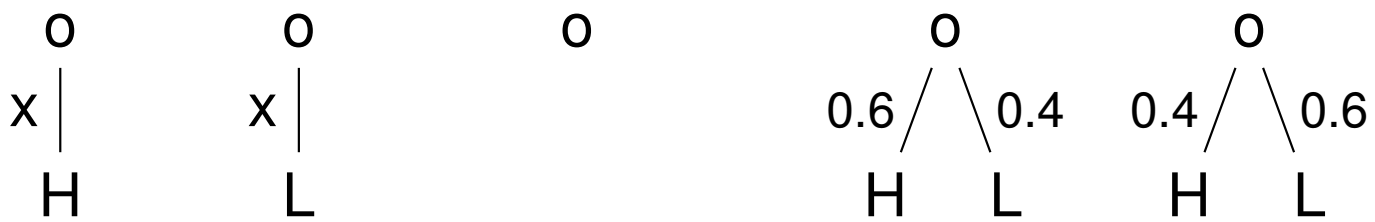
- Our analysis is couched into **Stratal Optimality Theory** with **gradient representations** of associations (Goldrick and Smolensky 2016, Kiparsky 2015, Bermúdez-Otero 2012).
- The analysis describes the **word-level phonology**. We assume that contour simplification (ML → M and HL → H) and neutralization of L˚ to L in non-final position happen later in the phrasal phonology.
- Optimization at some lower stratum ensures that values of association are **either 0, 0.4, 0.6, or 1.0** and nothing else.

Background: Representations

- **Mid tones and unreleased low tones** are represented as **mixed tones** on one tonal root node with different portions of activation of high and low tones. (similar to RTT cf. (Snider 1988))
- For high and low tones the activation is not relevant for the output interpretation.

(6) Representations of the four tones in Oku

a. High b. Low b. Weak Mid c. Strong Mid e. Low°



Analysis

Spreading

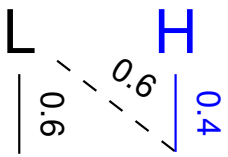
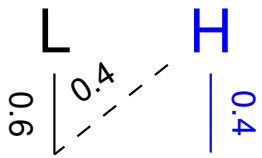
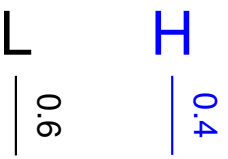
- **Spreading** is triggered by the constraint SPECIFY(T) (Myers 2004, Prince and Smolensky 1993) which requires every tonal root node to be associated by an activation of at least 1.0.
- The constraint SPEC(T)^{M} additionaly prefers spreading of **strong tones** over weak tones.

(7) Spreading trigger constraints

SPEC(T)	A TRN should be associated such that the sum of activations is at least 1.
SPEC(T)^{M}	A TRN should be associated such that the sum of activations is at least 0.5.

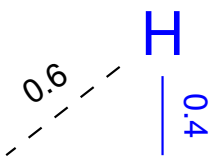
Low - Weak High

(8) Evaluation of a low tone stem plural

Input: = c.	*MIXSP	SPEC(T)	SPEC(T)^{M}
☞ a.  kà sə̀		*	
b.  kā sə́		*	*!
c.  kà sə́		**!	*

Weak Mid - Weak High

(9) Evaluation of a weak mid tone stem plural

Input: = c.	*MIXSP	SPEC(T)	SPEC(T)^{M}
☞ a. sə́ə sə́ 		**	*
b. səə sə 		**	**!
c. səə sə́ 		**	**!

Blocked Spreading

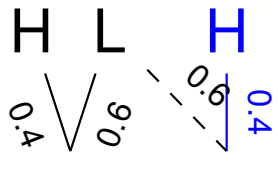
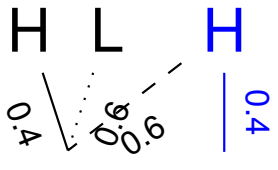
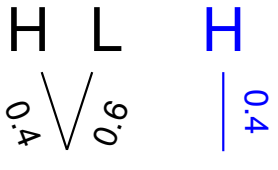
- *MIXSP **blocks** parts of the **mixed tones** from spreading.
- Sponsor: the underlying correspondent of an autosegmental span on the surface (cf. ODT (Cole and Kisseberth 1994))
- ASSOCIATE(Tone) **keeps** tones **from spreading** to the TRN that already is associated **to a mixed tone**.

(10)

*MIXEDSPONSOR	A sponsor TRN for tone τ should not be specified for tone τ' ($\tau' \neq \tau$)
ASSOC(T)	A tone should be associated to a TRN.

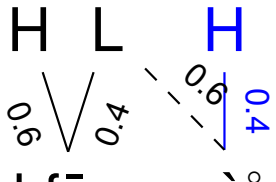
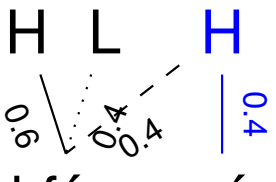
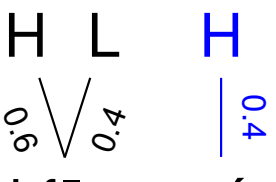
Low° - Weak High

(11) Evaluation of an unreleased low tone stem plural

Input: = c.	*MIXSP	SPEC(T)	ASSOC(T)
 a. njàm° sə̀°	*!		
 b. njám° sə́		*	*!
 ☞ c. njàm° sə́		*	

Strong Mid - Weak High

(12) Evaluation of a strong mid tone stem plural

Input: = c.	*MIXSP	SPEC(T)	ASSOC(T)
a.  kfən sə̀°	*!	*	
b.  kfə́n sə́		*	*!
☞ c.  kfə́n sə́		*	

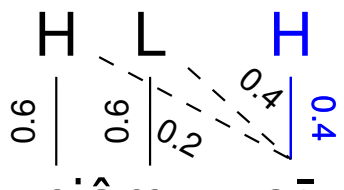
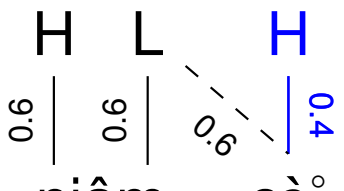
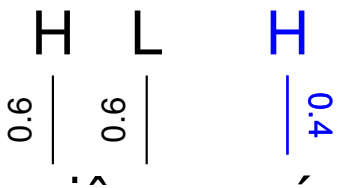
HL vs ML

- The low tone part of the contour tone spreads in both cases.
- The constraint *SPREAD^{M} prefers **spreading of both components** of the contour tone.
- *MIXSP blocks parts of the mid tone to spread to the suffix.

*SPREAD^{M} Do not insert an association line with an activation of more than 0.5.

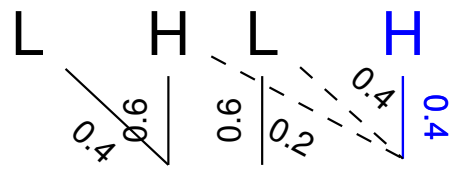
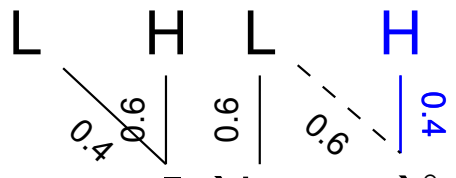
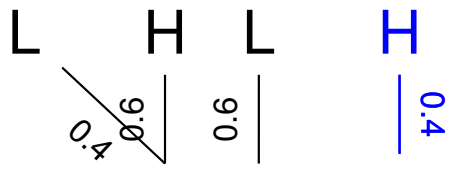
HL - Weak High

(13) Evaluation of a falling stem plural

Input: = c.	*MIXSP	SPEC(T)	*SPREAD^{M}
☞ a. 		**	
b. 		**	*!
c. 		* *! *	

ML - Weak High

(14) Evaluation of a mid low tone stem plural

Input: = c.	*MIXSP	SPEC(T)	*SPREAD^{M}
a. 	*!	*	*
☞ b. 		*	
c. 		* *!	

Conclusion

- We presented an account of Oku Tone Sandhi based on **gradient representations** of tone.
- This account deals with the patterns **without** making reference to **floating tones** as triggering or blocking phonological processes.
- Future work should concern a less abstract analysis of similar patterns amongst the world's languages.

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